TrES-2b

R-0090

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_180530` · created 2026-07-15T22:29:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458268.85489 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.018
Transit depth (Rp/Rs)^2	1.92 +/- 0.49 [%]
Orbital Inclination (inc)	84.2 +/- 1.4
Airmass coefficient 1 (a1)	1.04 +/- 0.016
Airmass coefficient 2 (a2)	-0.038 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	#3 - [119, 286]
Optimal Aperture	3.55
Optimal Annulus	5.26
Transit Duration (day)	0.075 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.018 vs 0.1254 (deviation 0.64σ)
Duration fit vs published	0.075 d vs 0.0746 d (deviation 0.01σ)
Mid-time O–C	-1.4 min
Depth SNR	7.0
Residual scatter	1.58 %

Light curve

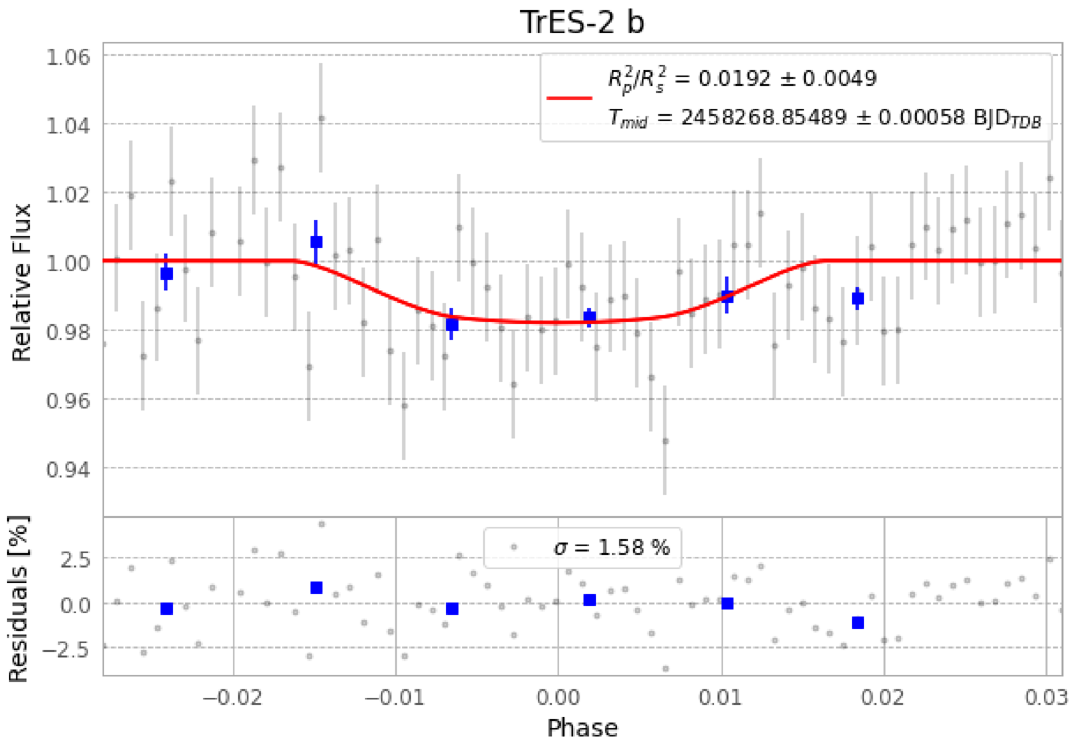

Figure 1 — FinalLightCurve_TrES-2 b_2018-05-29.png (SHA-256 acb977632e519810…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 242], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f3994084ac500770…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

71 FITS frames (47 MB), `TRES-21805300648l2.FITS` → `TRES-2180530101812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180530.log (efd7da10eb06…), FinalLightCurve_TrES-2 b_2018-05-29.png (acb977632e51…), FinalLightCurve_TrES-2 b_2018-05-29.csv (bde29c0a4b7c…)

Compute resources

Wall time 135.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)